S6 CHEMISTRY REVISION JAN 2021

IONIC EQUILIBRIA

1. (a) The table below shows acid dissociation constants K_a, for some acids at 25°C.

Acid	K_a(mol dm^{-3})
$H-COOH$	1.70×10^{-4}
CH_3COOH	1.70×10^{-5}
CH_3CH_2COOH	1.350×10^{-5}

(i) State the trend in acid strengths of the acids in the table above.

(ii) Explain your answer in a(i) above.

(b) Calculate the pH of a 0.5M CH_3CH_2COOH solution.

(c) (i) 45.0 cm^3 of a solution in (b) was mixed with 35.0 cm^3 of a 0.5M potassium hydroxide solution. Calculate the change in pH of the solution.

(ii) Predict the effect of adding two drops of dilute hydrochloric acid to the solution in c(i) above.

2. (a) Define the term buffer solution

(b) Explain the action of an acidic buffer

(c) 50 cm^3 of a 0.05M ethanoic acid was added to 50 cm^3 of a 0.02M sodium hydroxide solution.

(i) Write an equation for the reaction that took place

(ii) Calculate the pH of the resultant solution at 25°C. *(the acid dissociation constant K_a of ethanoic acid is $1.8\times 10^{-5} moldm^{-3}$ at 25°C.)*

(d) To 50 cm^3 of 0.02M ethanoic acid was added an equal volume of a 0.02M potassium hydroxide solution.

(i) Calculate the pH of the resultant solution

(ii) Explain your answer in d(i)

3. (a) Ammonium chloride solution gives effervescence with magnesium ribbon, whereas sodium ethanoate gives a blue precipitate with aqueous copper (II) chloride solution. Explain the observation.

(b) The hydrolysis constant of sodium methanoate is $6.25\times 10^{-11} moldm^{-3}$ at 25°C.

(i) Calculate the pH of a 0.1M aqueous solution of sodium methanoate solution at 25°C. *(the ionic product of water is $10^{-14} moldm^{-3}$ at 25°C.)*

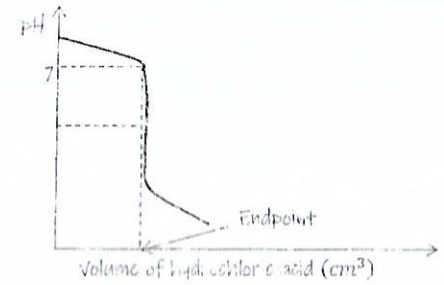
(ii) To 1 dm^3 of 0.1M sodium methanoate solution was added 10 cm^3 of 1M hydrochloric acid. Calculate the pH of the resultant solution. The acid dissociation constant of methanoic acid is $1.6\times 10^{-4} moldm^{-3}$.)

(c) Explain what would happen to the pH of a solution containing sodium methanoate and methanoic acid if it was added a small amount of

(i) hydrochloric acid

(ii) sodium hydroxide

4. (a)



The graph above shows the variation in pH of ammonia solution when it was titrated with dilute hydrochloric acid. Explain the shape of the graph.

(b) 1.07 g of ammonium chloride was dissolved in 1 dm^3 of 0.01M ammonia solution. Calculate the pH of the solution and state any two assumptions made in the calculation. (K_b for ammonia is = 1.78×10^{-5}moldm^{-3} and Kw = $1\times 10^{-14} mol^2dm^{-6}$)

(c)(i) Write the equation for the hydrolysis of sodium ethanoate in aqueous solution

(ii) Write an expression for the hydrolysis constant, K_h, of sodium ethanoate.

(d) The pH of a 0.1M aqueous sodium ethanoate solution is 8.9. Calculate the hydrolysis constant of the solution. ($K_w = 1\times 10^{-14} mol^2dm^{-6}$)

5. (a) (i) Define the terms **solubility product** and **common ion effect.**

(ii) State the factors that affect the value of solubility product

(b) Describe an experiment which can be carried out to determine the solubility product of

(i) calcium iodate, $Ca(IO_3)_2$

(ii) silver oxalate, $Ag_2C_2O_4$

(c) A saturated solution of silver oxalate contains 1.1×10^{-4} moles of silver oxalate per litre at 25°C. Calculate the solubility product of silver oxalate at 25°C and indicate its units.

(d) Determine the solubility of silver oxalate in 0.01M silver nitrate solution. State any assumptions you have made.